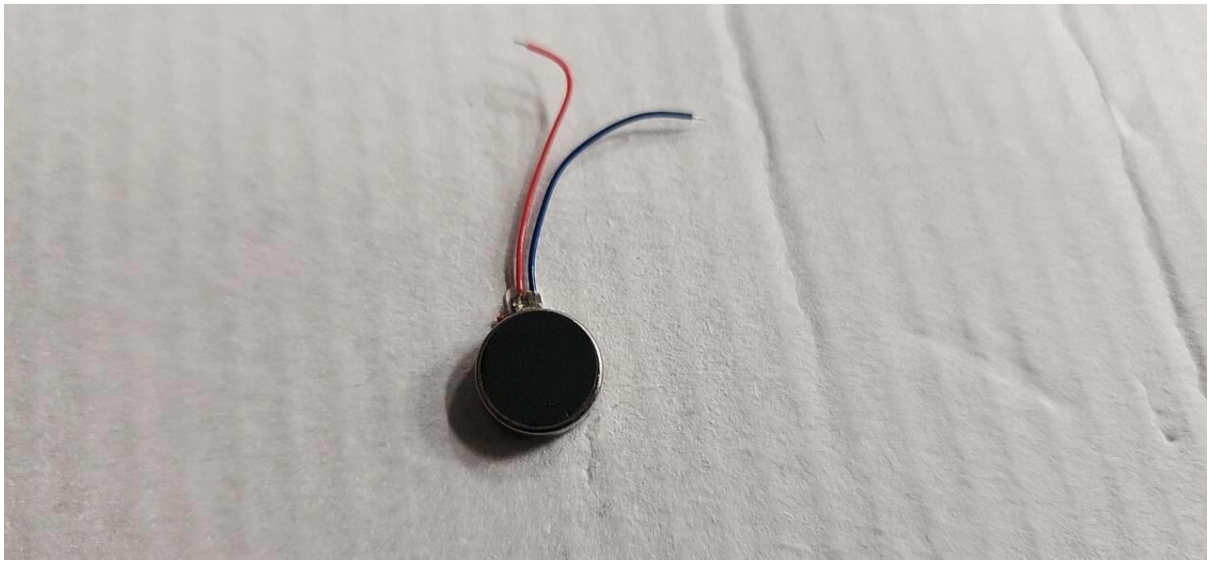
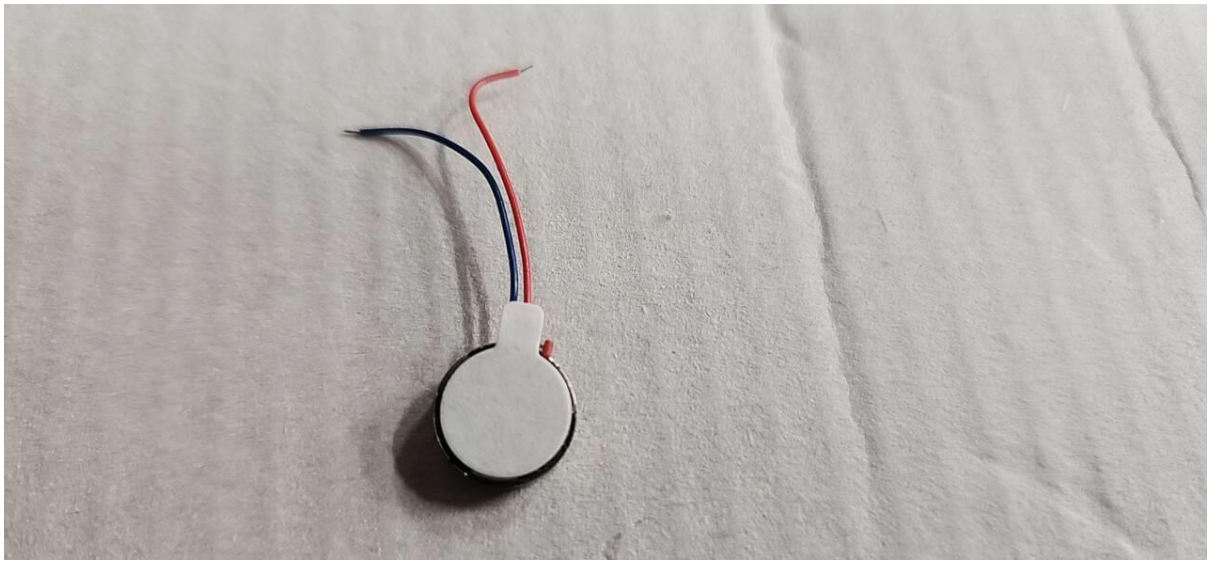
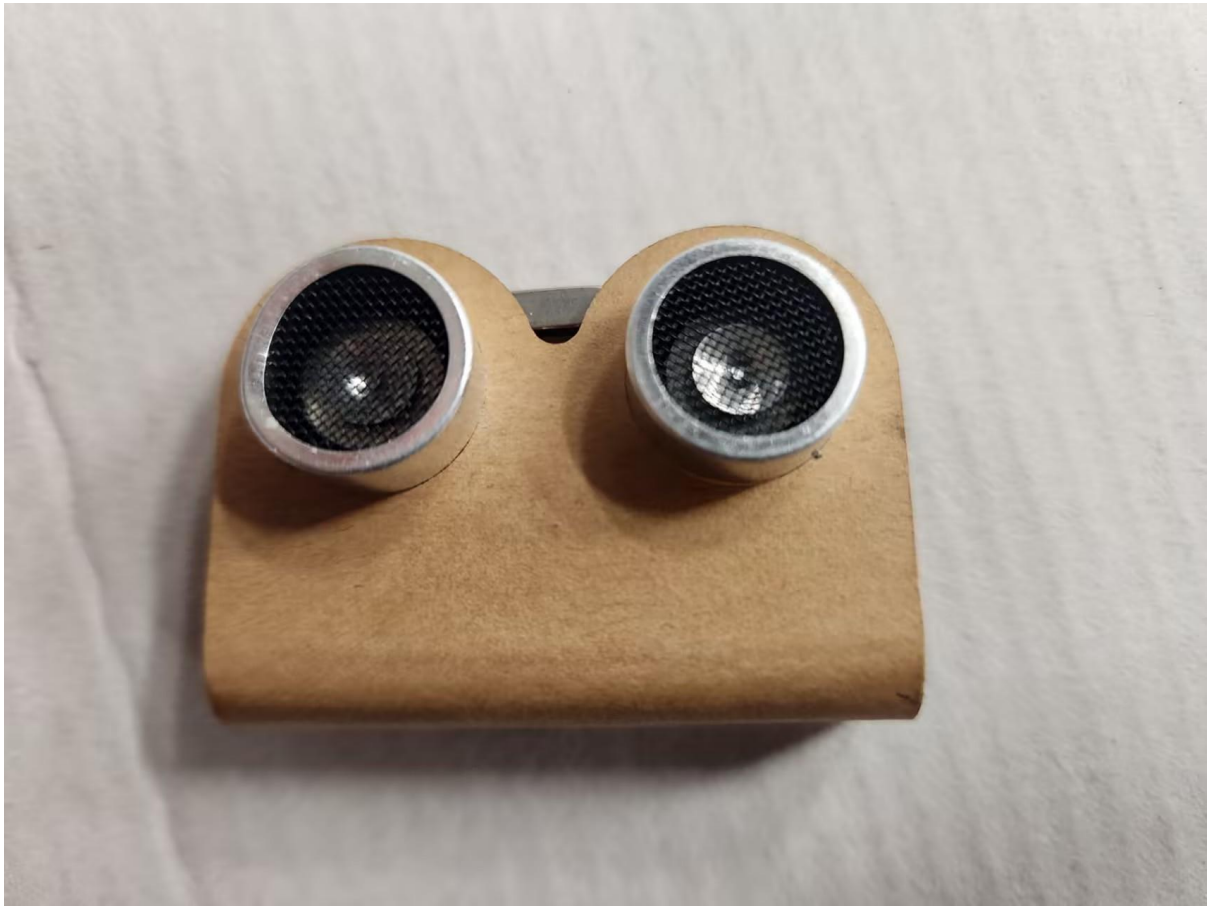
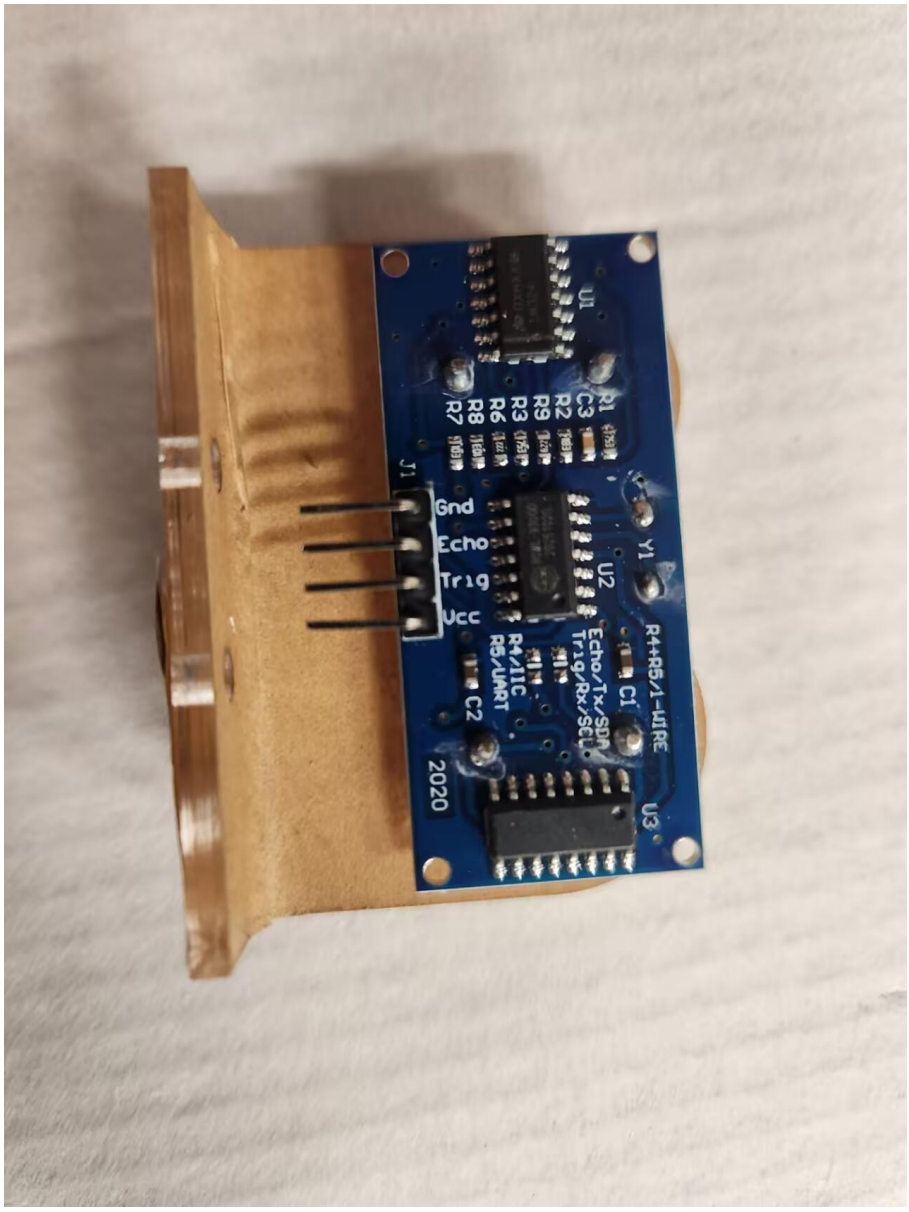
CS-492project





This is the front and back sides of the motor. The working voltage of this motor is 3.3 volts DC, but the working current is 80 to 120 mA. To protect the circuit, components such as transistors need to be connected; otherwise, there is a risk of circuit breakdown.



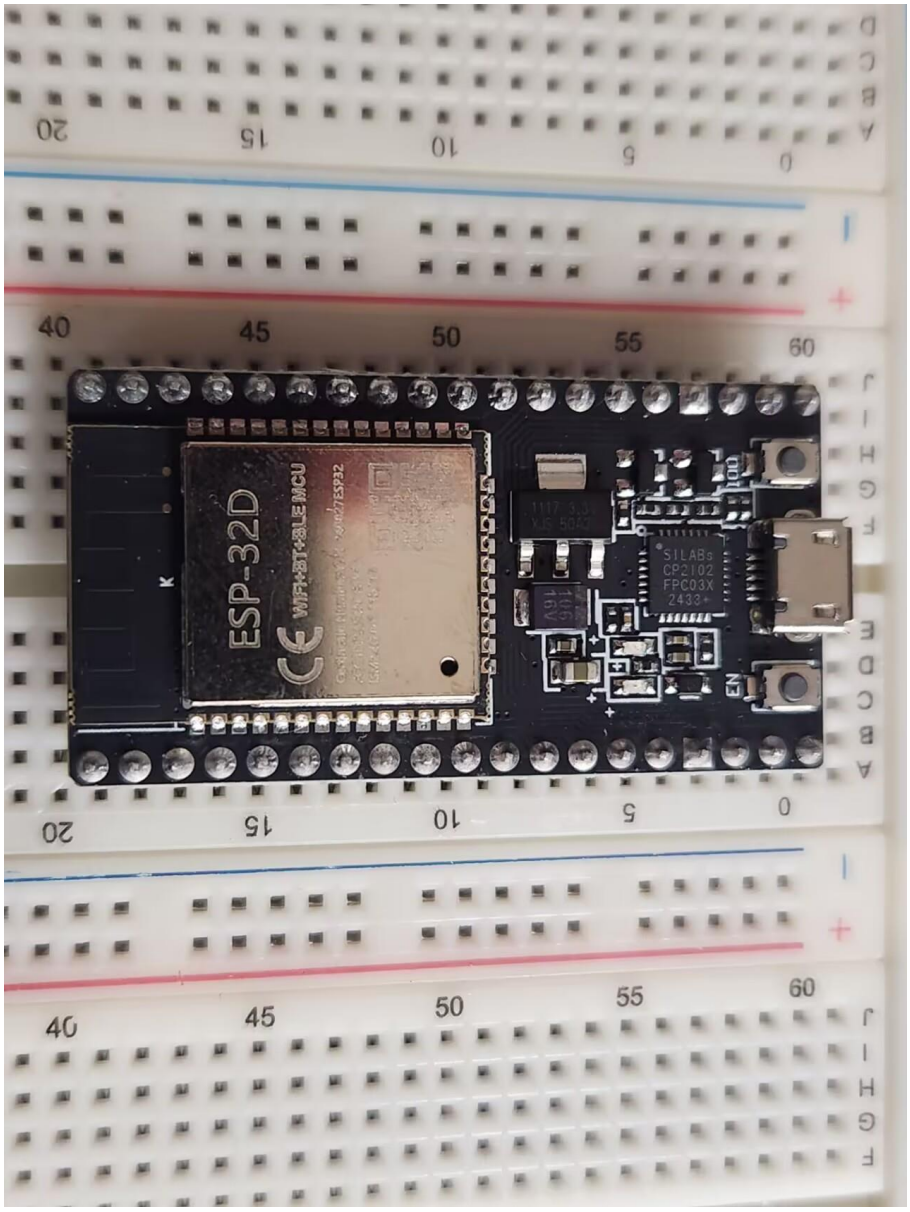


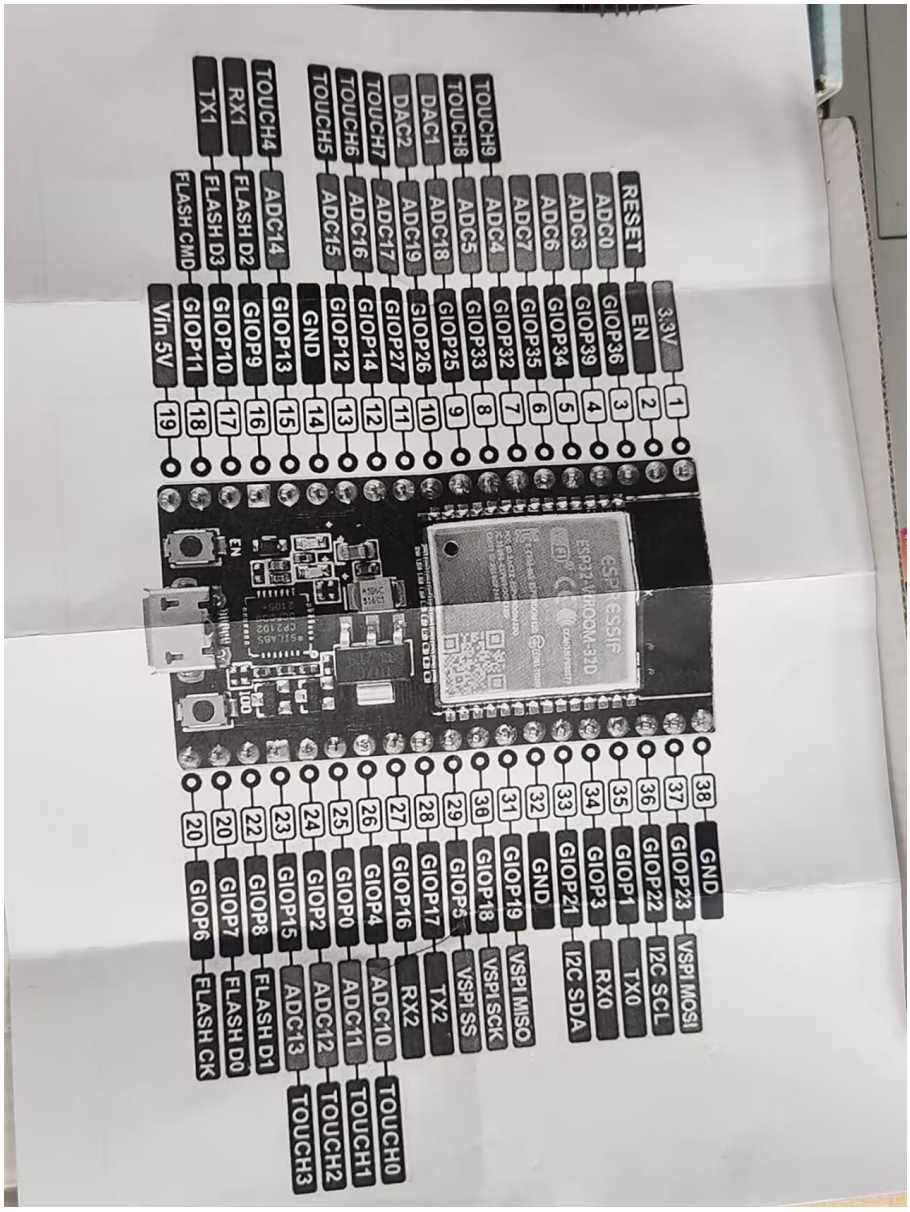
This is the HC-SR04 ultrasonic detector. Its working voltage is 5V and the working current is approximately 30mA. This can be directly connected to the ESP32 circuit board and does not require additional circuits for buffering and protection.


102
GND
I/O
VCC
低电平触发



This is a passive buzzer module. Its main function is to emit warning sounds. The operating voltage ranges from 3.3V to 5V, and the working current is approximately 30mA. It can be directly connected to the circuit.


ESP-32D
WiFi+BT+BLE MCU
SILABs
CP2102
FPC03X
2433+
EN



This is the main brain module of the circuit, ESP32. The voltage is 5V (the internal working voltage of the chip is around 3.3V). The working current is 100 to 250mA. It can be powered by a power bank, and the maximum output current of the pins is 12 to 40mA.